Difference-in-discontinuities Design*

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Abstract

This paper formalizes the difference-in-discontinuities design (shortly, diff-in-disc or RD-DID), a hybrid approach that combines the regression discontinuity design and the difference-in-difference design. While this design is widely employed in empirical works to mitigate the effects of confounding treatments co-occurring at discontinuities, few studies provide complete presentations of identifying assumptions and identified treatment effects. First, I provide a formalization of the difference-in-discontinuities design described by Grembi et al. (2016). Second, I clarify the identification framework for the difference-in-difference-in-discontinuities design. Notably, under the key identifying assumptions, the estimation design locally identifies the average treatment effect on the treated at the discontinuity, contrary to the proposition by Grembi et al. (2016) regarding the average treatment effect.

Keywords: Difference-in-discontinuities, RD-DD, RD-DDD, RD-DID, Regression discontinuity

JEL Codes: C10, C18, C21

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1 Introduction

The difference-in-discontinuities (shortly, diff-in-disc, RD-DID) design represents an expansion of the regression discontinuity design, incorporating a difference-in-differences framework to address confounding treatments at the discontinuity. In empirical settings, it is very common for the treatment of interest to coincide with other confounding events at the discontinuity, making this approach extensively used across various economics disciplines[1]. Furthermore, as a logical progression, empirical researchers have adopted the difference-in-difference-in-discontinuities (shortly, diff-in-diff-in-disc or RD-DDD) design, which combines the regression discontinuity design with a triple difference framework to further mitigate confounding treatment effects (Neugart and Ohlsson, 2013; Danzer and Lavy, 2018; Pless and Srivastav, 2022; Antwi et al., 2023). Additionally, the diff-in-disc design finds increasing application in varied contexts, such as fuzzy discontinuity (Millán-Quijano, 2020; Larsen and Valant, 2023) and geographic discontinuity (Butts, 2023).

Despite its popularity and potential to address issues in the regression discontinuity design, the standard diff-in-disc design has not been fully formalized in econometric literature or causal inference textbooks, such as those by Angrist and Pischke (2009) and Cunningham (2021). Notably, Grembi et al. (2016) is the first to explicitly state the identifying assumptions and the identified treatment effect of the diff-in-disc design in their empirical setting. However, other applications have primarily provided only verbal explanations of the identification strategy. Regarding the diff-in-diff-in-disc estimator, to the best of my knowledge, there are no studies that offer a complete identification framework. With an increasing number of studies employing the diff-in-disc design, there is a clear need for research that systematically presents the identifying assumptions and the identified treatment effect in a

[1] Refer to the next section and the appendix for a list of the top 50 most cited articles employing the diff-in-disc design.

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clear and explicit manner.

This paper presents formal representations of diff-in-disc and diff-in-diff-in-disc designs and derives identifying assumptions and identified treatment effects that are not clearly stated in many applications. To facilitate a deeper understanding, I examine examples of policy changes as leveraged by Grembi et al. (2016) and Danzer and Lavy (2018), with the goal of elucidating the types of confounding treatments that the diff-in-disc design addresses. Crucially, I assert that, under the key identifying assumptions, these estimation designs locally identify the average treatment effect on the treated at the discontinuity. This assertion contradicts the argument put forth by Grembi et al. (2016), which claims that the estimated effect is the average treatment effect at the discontinuity. The distinction in the identified treatment effect is pivotal because it can potentially change the policy implication.

The remainder of the paper is as follows: Section 2 reviews the literature. Section 3 formalizes the diff-in-disc design as explained by Grembi et al. (2016). Section 4 provides an exhaustive representation of the identifying assumptions and identified effects for the diff-in-diff-in-disc design. Section 5 discuss issues related to estimation. Section 6 concludes.

2 Literature

The origin of the diff-in-disc design is unclear because the author may not explicitly name the identification method while using it. Even in cases where it is named, there are various labels, including major ones like "diff-in-disc" and "RD-DID," as well as alternative labels such as "DDisc," "DID-RD," "D-DD," "RD-DD," and "RD-DiD." To the best of my knowledge, Dickert-Conlin and Elder (2010) is the earliest paper that utilized the diff-in-disc design, while Neugart and Ohlsson (2013) is the first paper to introduce the diff-in-diff-in-disc design. Importantly, Grembi et al. (2016)[2]

[2]There is a working paper version Grembi et al. (2012).

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first explicitly represent the identifying assumptions and the identified treatment effects under their empirical setting, paving the way for subsequent applications.

To demonstrate the significance of the diff-in-disc design in economics (and beyond), I conduct a literature review of the most frequently cited articles utilizing this identification design. I identified the top 50 most cited published articles[3] through a Google Scholar search. Table A.1 in the appendix displays the list of articles.

First, the result reveals widespread use of the diff-in-disc design in economics. Notably, articles in high-impact general interest journals such as the American Economic Review, Econometrica, and the Journal of Political Economy employ this design. Moreover, papers utilizing this identification design are accepted in top journals across various fields. Specifically, the design is prevalent in development, health, labor, public, and corporate finance, suggesting its effectiveness in program evaluation contexts. Additionally, political science frequently employs the design.

Secondly, the list of articles also underscores the impact of Grembi et al. (2016) in popularizing the method within the research community. An investigation into the references used for the identification strategy reveals that 48% (24/50) of the papers cite either Grembi et al. (2012) or Grembi et al. (2016). However, approximately half of the papers do not reference a specific source, and the nomenclature used is not standardized, as previously noted. Furthermore, the argument proposed by Grembi et al. (2016) lacks formalization and contains a minor error (discussed in the following section). These observations underscore the significance of and the need for further elucidation of the diff-in-disc design. Consequently, I provide a comprehensive representation of the design in subsequent sections.

[3] As of January 15, 2024.

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3 Difference-in-discontinuities

I first provide a formalization of the diff-in-disc design, which is firstly described by Grembi et al. (2016). In this context, researchers are aware that the treatment of interest, denoted as D_i, is triggered by the discontinuity $X_i = c$. However, there exists a confounding treatment V_i that is also triggered by the discontinuity. While both treatments share the same discontinuity threshold, the treatment of interest, D_i, is exclusively applied within the specific subgroup $T_i = 1$. This setting is realistic in the context of policy evaluation. Before presenting the identifying assumptions, I show an empirical example.

Example 1. *Grembi et al. (2016) investigate the effect of the fiscal policy on municipality outcomes. In 2001, fiscal rules were relaxed for municipalities with below 5,000 inhabitants in Italy. Though below/above 5,000 seems a standard regression discontinuity cutoff, there is another policy that the mayor's salary sharply changes at the 5,000 inhabitants threshold. In this context, before/after 2001 forms comparable groups to cancel out the confounding effect of the mayor's salary treatment.*

The treatments are defined as:

$$V_i = \begin{cases} 1 & \text{if } X_i \geq c \\ 0 & \text{otherwise} \end{cases}$$

$$D_i = \begin{cases} 1 & \text{if } X_i \geq c \text{ and } T_i = 1 \\ 0 & \text{otherwise} \end{cases}$$

Suppose the potential outcome is $Y_i(V_i = v, D_i = d)$. Therefore, the observed outcome can be written as:

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$$
\begin{aligned}
Y_i &= (1-V_i)(1-D_i)Y_i(0,0) + V_i(1-D_i)Y_i(1,0) \\
&\quad + (1-V_i)D_iY_i(0,1) + V_iD_iY_i(1,1)
\end{aligned}
$$

Note that the estimable causal effect of the treatment of interest D_i is $Y_i(1,1) - Y_i(1,0)$ under this potential outcome framework. One possible regression-discontinuity estimator is the mean difference in outcomes for observations above and below the discontinuity (Imbens and Lemieux, 2008). The identified treatment effect of the regression-discontinuity design using only the treated sample ($T_i = 1$) can be written as:

$$
\begin{aligned}
\tau_{RD|T=1} &\equiv \lim_{x\to c^+} E[Y_i|X_i = x, T_i = 1] - \lim_{x\to c^-} E[Y_i|X_i = x, T_i = 1] \\
&= \lim_{x\to c^+} E[Y_i(1,1)|X_i = x, T_i = 1] - \lim_{x\to c^-} E[Y_i(0,0)|X_i = x, T_i = 1] \\
&= E[\underbrace{Y_i(1,1) - Y_i(1,0)}_{\text{causal effect of } D_i}|X_i = c, T_i = 1] \\
&\quad + E[\underbrace{Y_i(1,0) - Y_i(0,0)}_{\text{causal effect of } V_i}|X_i = c, T_i = 1]
\end{aligned}
$$

Importantly, the identified treatment effect is biased, and it can be decomposed into the causal effect of D_i and the causal effect of V_i. As mentioned earlier, the causal effect of V_i can be eliminated by exploiting the control sample ($T_i = 0$). Under the diff-in-disc design, the identified treatment effect can be defined as follows:

$$
\tau_{DD} \equiv \tau_{RD|T=1} - \tau_{RD|T=0} \tag{1}
$$

To identify the causal effect of D_i, the following assumptions are made:

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Assumption 1. *All potential outcomes* $E[Y_i(v,d)|X_i = x, T_i = t]$ *for* $(v,d) \in \{(0,0),(1,0),(1,1)\}$*, and* $t \in \{0,1\}$ *are continuous in* x *at* c

Assumption 2. *The effects of the confounding treatment* V_i *at the discontinuity* c *are constant across the groups:* $E[Y_i(1,0) - Y_i(0,0)|X_i = c, T_i = 1] = E[Y_i(1,0) - Y_i(0,0)|X_i = c, T_i = 0]$

Assumption 1 is a standard and minimally required assumption of continuity in regression-discontinuity literature (Imbens and Lemieux, 2008; Cunningham, 2021). Assumption 2 is analogous to the parallel trend assumption in the difference-in-difference design, but it is more localized and called a local parallel trend borrowed from Grembi et al. (2016). The assumption requires that two comparable groups have a constant jump at the discontinuity due to the confounding treatments. Consequently, the following proposition can be proposed:

Proposition 1. *Under Assumption 1 and Assumption 2, the diff-in-disc design locally identifies the average treatment effect on the treated of* D_i*:* $E[Y_i(1,1) - Y_i(1,0)|X_i = c, T_i = 1]$

Proof.

$$\tau_{RD|T=0} \equiv E[Y_i(1,0) - Y_i(0,0)|X_i = c, T_i = 0]$$

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$$
\begin{aligned}
\tau_{DD} &\equiv E[Y_i(1,1) - Y_i(1,0)|X_i = c, T_i = 1] \\
&\quad + E[Y_i(1,0) - Y_i(0,0)|X_i = c, T_i = 1] \\
&\quad - E[Y_i(1,0) - Y_i(0,0)|X_i = c, T_i = 0] \\
&= E[Y_i(1,1) - Y_i(1,0)|X_i = c, T_i = 1]
\end{aligned}
$$

□

Note that the estimated treatment effect represents the average treatment effect on the treated (ATT) at the discontinuity. While Grembi et al. (2016) assert that the average treatment effect (ATE) is identified under the assumptions laid out in their proposition, an additional assumption regarding the random assignment of the group is necessary to actualize this. Some literature[4] also posits that the estimated effect corresponds to the ATE (Lopes da Fonseca, 2017; Butts, 2023), potentially stemming from the proposition by Grembi et al. (2016). This distinction is significant because the group for which the treatment effect is estimated can alter policy implications, although the magnitude of the effect may vary across empirical settings.

4 Difference-in-difference-in-discontinuities

To address the additional confounding treatment issues in the diff-in-disc design, I introduce the diff-in-diff-in-disc design, which has not been explicitly detailed in the literature. Besides the confounding treatment V_i triggered by the discontinuity $X_i = c$, consider another confounding treatment W_i triggered by the discontinuity within the treated group ($T_i = 1$). In the previous scenario, using the comparable

[4] Additionally, some literature asserts that the estimated effect constitutes the local average treatment effect (LATE) (Millán-Quijano, 2020; Laajaj et al., 2022). While this proposition is accurate to a degree, it remains ambiguous whether the aspect of the ATT is contemplated within the term LATE here.

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group based on T_i aided in mitigating the effects of the confounding treatment V_i. However, the new confounding treatment W_i cannot be effectively addressed by the existing diff-in-disc design. Suppose there is a new group B_i, and the treatment of interest D_i is exclusively triggered by the discontinuity within this specific subgroup $T_i = 1$ and $B_i = 1$. The introduction of this new source of treatment assignments can act as a critical identifying tool within the triple difference estimator framework, particularly relevant in policy evaluation contexts, especially those utilizing date-based discontinuities. Before delving into the identification framework, I present an empirical example to illustrate this concept.

Example 2. *Danzer and Lavy (2018) examined the impact of changes in paid maternity leave policy on the future academic outcomes of children. On July 1, 1990, the duration of paid parental leave was extended in Austria. While utilizing the before/after July 1, the beginning of the month, as a discontinuity could potentially introduce confounding systematic differences, exploiting before/after 1990 can cancel out the bias specific to July 1. However, there may be numerous potential confounding treatment events occurring exclusively on July 1, 1990, which cannot be addressed through the diff-in-disc design alone. Applying the triple difference framework, the country-level difference between Austria/Germany creates comparable groups to reduce the effects of confounding treatments specific to July 1, 1990.*

The treatments are defined as:

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$$
\begin{aligned}
V_i &= \begin{cases} 1 & \text{if } X_i \geq c \\ 0 & \text{otherwise} \end{cases} \\
W_i &= \begin{cases} 1 & \text{if } X_i \geq c \text{ and } T_i = 1 \\ 0 & \text{otherwise} \end{cases} \\
D_i &= \begin{cases} 1 & \text{if } X_i \geq c,\ T_i = 1, \text{ and } B_i = 1 \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
$$

The potential outcome is $Y_i(V_i = v, W_i = w, D_i = d)$. Therefore, the observed outcome can be written as:

$$
\begin{aligned}
Y_i = \; & (1 - V_i)(1 - W_i)(1 - D_i)Y_i(0,0,0) + V_i(1 - W_i)(1 - D_i)Y_i(1,0,0) \\
& + (1 - V_i)W_i(1 - D_i)Y_i(0,1,0) + (1 - V_i)(1 - W_i)D_iY_i(0,0,1) \\
& + V_iW_i(1 - D_i)Y_i(1,1,0) + V_i(1 - W_i)D_iY_i(1,0,1) \\
& + (1 - V_i)W_iD_iY_i(0,1,1) + V_iW_iD_iY_i(1,1,1)
\end{aligned}
$$

Note that the estimable causal effect of the treatment of interest D_i is $Y_i(1,1,1) - Y_i(1,1,0)$. Assumption 1 and Assumption 2 can be updated as follows:

Assumption 3. *All potential outcomes* $E[Y_i(v,w,d)|X_i = x, T_i = t, B_i = b]$ *for* $(v,w,d) \in \{(0,0,0),(1,0,0),(1,1,0),(1,1,1)\}$, $t \in \{0,1\}$, *and* $b \in \{0,1\}$ *are continuous in* x *at* c

Assumption 4. *The effects of the confounding treatment* V_i *at the discontinuity* c *are constant across the first group:* $E[Y_i(1,0,0) - Y_i(0,0,0)|X_i = c, T_i = 1] =$

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$E[Y_i(1,0,0) - Y_i(0,0,0)|X_i = c, T_i = 0]$

Under Assumption 3 and Assumption 4, the treatment effect identified by applying the diff-in-disc design to one subgroup ($B_i = 1$) can be written as:

$$\begin{aligned}
\tau_{DD|B=1} &\equiv \tau_{RD|T=1,B=1} - \tau_{RD|T=0,B=1} \\
&= E[Y_i(1,1,1) - Y_i(0,0,0)|X_i = c, T_i = 1, B_i = 1] \\
&\quad - E[Y_i(1,0,0) - Y_i(0,0,0)|X_i = c, T_i = 0, B_i = 1] \\
&= E[Y_i(1,1,1) - Y_i(1,1,0)|X_i = c, T_i = 1, B_i = 1] \\
&\quad + E[Y_i(1,1,0) - Y_i(1,0,0)|X_i = c, T_i = 1, B_i = 1] \\
&\quad + E[Y_i(1,0,0) - Y_i(0,0,0)|X_i = c, T_i = 1, B_i = 1] \\
&\quad - E[Y_i(1,0,0) - Y_i(0,0,0)|X_i = c, T_i = 0, B_i = 1] \\
&= E[\underbrace{Y_i(1,1,1) - Y_i(1,1,0)}_{\text{causal effect of } D_i}|X_i = c, T_i = 1, B_i = 1] \\
&\quad + E[\underbrace{Y_i(1,1,0) - Y_i(1,0,0)}_{\text{causal effect of } W_i}|X_i = c, T_i = 1, B_i = 1]
\end{aligned}$$

The diff-in-disc design no longer captures solely the causal effect of the treatment of interest D_i but also the causal effect of the confounding treatment W_i. The diff-in-diff-in-disc designed to eliminate the bias arising from the confounding treatment W_i. By leveraging the variation between two groups, the diff-in-diff-in-disc estimator effectively represents the difference between the diff-in-disc estimators. The identified treatment effect can be defined as follows:

$$\tau_{DDD} \equiv \tau_{DD|B=1} - \tau_{DD|B=0} \tag{2}$$

To identify the causal effect of D_i, the following additional assumption is made:

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Assumption 5. *The effects of the confounding treatment W_i at the discontinuity c on the first treated group are constant across the second group:* $E[Y_i(1,1,0) - Y_i(1,0,0)|X_i = c, T_i = 1, B_i = 1] = E[Y_i(1,1,0) - Y_i(1,0,0)|X_i = c, T_i = 1, B_i = 0]$

Assumption 5 represents the local parallel trend assumption for the confounding treatment W_i among the two group based on B_i, similar to Assumption 4. However, the local parallel trend requirements for the two groups are not precisely equivalent because the required local parallel trend assumption for B_i is solely based on the subgroup where $T_i = 1$. This is analogous to the notion that two parallel trends are not necessary for two groups in the triple difference design (Olden and Møen, 2022). Consequently, the following proposition can be articulated:

Proposition 2. *Under Assumption 3, Assumption 4, and Assumption 5, the diff-in-diff-in-disc design identifies the average treatment effect on the treated of D_i:* $E[Y_i(1,1,1) - Y_i(1,1,0)|X_i = c, T_i = 1, B_i = 1]$

Proof.

$$
\begin{aligned}
\tau_{DD|B=0} &\equiv \tau_{RD|T=1,B=0} - \tau_{RD|T=0,B=0} \\
&= E[Y_i(1,1,0) - Y_i(0,0,0)|X_i = c, T_i = 1, B_i = 0] \\
&\quad - E[Y_i(1,0,0) - Y_i(0,0,0)|X_i = c, T_i = 0, B_i = 0] \\
&= E[Y_i(1,1,0) - Y_i(1,0,0)|X_i = c, T_i = 1, B_i = 0] \\
&\quad + E[Y_i(1,0,0) - Y_i(0,0,0)|X_i = c, T_i = 1, B_i = 0] \\
&\quad - E[Y_i(1,0,0) - Y_i(0,0,0)|X_i = c, T_i = 0, B_i = 0] \\
&= E[Y_i(1,1,0) - Y_i(1,0,0)|X_i = c, T_i = 1, B_i = 0]
\end{aligned}
$$

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$$
\begin{aligned}
\tau_{DDD} &\equiv E[Y_i(1,1,1) - Y_i(1,1,0)|X_i = c, T_i = 1, B_i = 1] \\
&\quad + E[Y_i(1,1,0) - Y_i(1,0,0)|X_i = c, T_i = 1, B_i = 1] \\
&\quad - E[Y_i(1,1,0) - Y_i(1,0,0)|X_i = c, T_i = 1, B_i = 0] \\
&= E[Y_i(1,1,1) - Y_i(1,1,0)|X_i = c, T_i = 1, B_i = 1]
\end{aligned}
$$

□

5 Estimation

The estimation is primarily implemented using local polynomial methods, which fit the outcomes to a low-order polynomial expansion of the running variable. This fitting is done separately for each treatment status of observations within a distance h on either side of the discontinuity. For the diff-in-disc case, four regressions are estimated, while eight regressions are estimated for the diff-in-diff-in-disc case. In the application literature, local linear regression is the popular choice. Importantly, the regression specification fits a difference-in-difference style, including interaction terms. By restricting the sample to observations with $X_i \in [c-h, c+h]$, the regression function can be estimated by solving the following problem:

$$
\min_{\beta_0,\ldots\beta_3} \sum_i (Y_i - \beta_0 - \beta_1 \mathbb{1}(X_i \geq c) - \beta_2 T_i - \beta_3 \mathbb{1}(X_i \geq c) \cdot T_i)^2
$$

The interpretation of the coefficients corresponds to the difference-in-difference specification, with the coefficient β_3 being the diff-in-disc estimator. To complete the estimation, the bandwidths h need to be determined, which are often optimally computed through algorithms developed by Imbens and Kalyanaraman (2012) and

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Calonico et al. (2014) in the literature[5].

6 Conclusion

The diff-in-disc design provides a regression-discontinuity-based approach for identifying treatment effects in scenarios where confounding treatments are present at the discontinuity. Incorporating a difference-in-differences approach aids in mitigating the effects of these confounding treatments, given the critical identifying assumption of local parallel trends at the discontinuity. It is crucial to recognize that this method primarily identifies the ATT at the discontinuity, highlighting its localized application. This specificity stems from the combined regression-discontinuity and difference-in-differences frameworks of the design.

This paper contributes in two significant ways, aiming to serve as a practical guide for the diff-in-disc design. Firstly, I offer the first exhaustive overview of the diff-in-disc design, demonstrating its popularity within the field of economics. Secondly, I formalize the diff-in-disc design, discovering that the identified effect is not the ATE but rather the ATT. Although the identified treatment effect is highly localized, the diff-in-disc design proves to be invaluable for addressing the critical identification challenges posed by concurrent events in the regression discontinuity framework. Empirical researchers are encouraged to apply this design, taking into account the treatment status in each empirical context.

[5]Note that many pieces of literature conduct robustness checks using different bandwidths regardless of using the optimal bandwidths.

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Table A.1: **Top 50 Most Cited Published Articles Using Difference-in-discontinuities Design**

Note: This table displays the top 50 most cited published articles that utilize the difference-in-discontinuities design, according to a survey based on Google Scholar search results as of January 15, 2024. The "Ref" column denotes whether the paper cites Grembi et al. (2016) or Grembi et al. (2012).

	Cites	Author	Title	Year	Ref	Journal
1	598	V Grembi, T Nannicini, U Troiano	Do fiscal rules matter?	2016	-	AEJ: Applied Economics
2	439	P Carneiro, KV Løken...	A flying start? Maternity leave benefits and long-run outcomes of children	2015	0	Journal of Political Economy
3	225	M Bennedsen, E Simintzi, M Tsoutsoura...	Do firms respond to gender pay gap transparency?	2022	1	Journal of Finance
4	192	MD Cattaneo, R Titiunik	Regression discontinuity designs	2022	1	Annual Review of Economics
5	190	AC Eggers, R Freier, V Grembi...	Regression discontinuity designs based on population thresholds: Pitfalls and solutions	2018	1	American Journal of Political Science
6	158	AC Smith	Spring forward at your own risk: Daylight saving time and fatal vehicle crashes	2016	0	AEJ: Applied Economics
7	144	M Deshpande	Does welfare inhibit success? The long-term effects of removing low-income youth from the disability rolls	2016	0	American Economic Review
8	139	PJ McEwan	The impact of Chile's school feeding program on education outcomes	2013	0	Economics of Education Review
9	135	M Ehrlich, T Seidel	The persistent effects of place-based policy: Evidence from the West-German Zonenrandgebiet	2018	0	AEJ: Economic Policy
10	127	JD Clinton, MW Sances	The politics of policy: The initial mass political effects of medicaid expansion in the states	2018	0	American Political Science Review
11	116	M Carlana, E La Ferrara, P Pinotti	Goals and gaps: Educational careers of immigrant children	2022	0	Econometrica
12	114	S Miller, LR Wherry, DG Foster	The economic consequences of being denied an abortion	2023	0	AEJ: Economic Policy
13	112	F Carta, L Rizzica	Early kindergarten, maternal labor supply and children's outcomes: evidence from Italy	2018	1	Journal of Public Economics
14	110	JM Jenkins, G Farkas, GJ Duncan...	Head Start at ages 3 and 4 versus Head Start followed by state pre-k: Which is more effective?	2016	1	Educational Evaluation and Policy Analysis
15	108	S Dickert	Suburban legend: School cutoff dates and the timing of births	2010	0	Economics of Education Review

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Table A.2: **Top 50 Most Cited Published Articles Using Difference-in-discontinuities Design (Continued)**

	Cites	Author	Title	Year	Ref	Journal
16	101	N Danzer, V Lavy	Paid parental leave and children's schooling outcomes	2018	0	The Economic Journal
17	97	D Avdic, A Karimi	Modern family? Paternity leave and marital stability	2018	0	AEJ: Applied Economics
18	88	M Ning, J Gong, X Zheng, J Zhuang	Does new rural pension scheme decrease elderly labor supply? Evidence from CHARLS	2016	0	China Economic Review
19	88	A Baltrunaite, A Casarico, P Profeta, G Savio	Let the voters choose women	2019	1	Journal of Public Economics
20	86	J Geyer, C Welteke	Closing routes to retirement for women: How do they respond?	2021	0	Journal of Human Resources
21	80	A Shenoy	Regional development through place-based policies: Evidence from a spatial discontinuity	2018	1	Journal of Development Economics
22	79	Z Asatryan, T Baskaran, T Grigoriadis...	Direct democracy and local public finances under cooperative federalism	2017	1	The Scandinavian Journal of Economics
23	79	JP Robinson-Cimpian...	The effects of changing test-based policies for reclassifying English learners	2016	0	Journal of Human Resources
24	75	A Hijzen, L Mondauto, S Scarpetta	The impact of employment protection on temporary employment: Evidence from a regression discontinuity design	2017	1	Labour Economics
25	73	J de Benedictis	Off-cycle and out of office: Election timing and the incumbency advantage	2018	1	The Journal of Politics
26	67	JD Gottlieb, RR Townsend, T Xu	Does career risk deter potential entrepreneurs?	2022	1	The Review of Financial Studies
27	65	A Bütikofer, J Riise, M M. Skira	The impact of paid maternity leave on maternal health	2021	0	AEJ: Economic Policy
28	61	H Li, J Li, Y Lu, H Xie	Housing wealth and labor supply: Evidence from a regression discontinuity design	2020	1	Journal of Public Economics
29	60	A Olafsson, H Steingrimsdottir	How does daddy at home affect marital stability?	2020	0	The Economic Journal
30	59	B Argyle, T Nadauld, C Palmer	Real effects of search frictions in consumer credit markets	2023	1	The Review of Financial Studies
31	55	A Kochar	Branchless banking: Evaluating the doorstep delivery of financial services in rural India	2018	1	Journal of Development Economics
32	54	A Nilsson, A Paul	Patient cost-sharing, socioeconomic status, and children's health care utilization	2018	0	Journal of Health Economics
33	53	R Laajaj, A Moya, F Sánchez	Equality of opportunity and human capital accumulation: Motivational effect of a nationwide scholarship in Colombia	2022	1	Journal of Development Economics

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Table A.3: **Top 50 Most Cited Published Articles Using Difference-in-discontinuities Design (Continued)**

	Cites	Author	Title	Year	Ref	Journal
34	51	G Khanna	Does affirmative action incentivize schooling? Evidence from India	2020	1	The Review of Economics and Statistics
35	46	M Duggan, A Gupta, E Jackson	The impact of the Affordable Care Act: evidence from California's hospital sector	2022	0	AEJ: Economic Policy
36	42	BR Frandsen	The surprising impacts of unionization: Evidence from matched employer-employee data	2021	1	Journal of Labor Economics
37	40	K Cygan	Paid parental leave and families' living arrangements	2018	0	Labour Economics
38	40	E Conover, D Scrimgeour	Health consequences of easier access to alcohol: New Zealand evidence	2013	1	Journal of Health Economics
39	39	M Lopes da Fonseca	Identifying the source of incumbency advantage through a constitutional reform	2017		American Journal of Political Science
40	38	F Carozzi, CAL Hilber, X Yu	On the economic impacts of mortgage credit expansion policies: evidence from help to buy	2024	0	Journal of Urban Economics
41	36	J Lassébie	Gender quotas and the selection of local politicians: Evidence from French municipal elections	2020	1	European Journal of Political Economy
42	35	L Marattin, T Nannicini, F Porcelli	Revenue vs expenditure based fiscal consolidation: the pass-through from federal cuts to local taxes	2022	1	International Tax and Public Finance
43	34	J Scott	The consequences of performance standards in need-based aid evidence from community colleges	2020	0	Journal of Human Resources
44	34	L González, SK Trommlerová	Cash transfers and fertility: How the introduction and cancellation of a child benefit affected births and abortions	2023	0	Journal of Human Resources
45	32	C Piza	Out of the Shadows? Revisiting the impact of the Brazilian SIMPLES program on firms' formalization rates	2018	0	Journal of Development Economics
46	32	TG Meling	Anonymous trading in equities	2021		Journal of Finance
47	32	M Picchio, R Santolini	Fiscal rules and budget forecast errors of Italian municipalities	2020	1	European Journal of Political Economy
48	32	M Hazan, D Weiss, H Zoabi	Women's liberation as a financial innovation	2019	1	Journal of Finance
49	31	HH Chang	Old farmer pension program and farm succession: evidence from a population-based survey of farm households in Taiwan	2013	0	American Journal of Agricultural Economics
50	31	RM Gonzalez	Cell phone access and election fraud: evidence from a spatial regression discontinuity design in Afghanistan	2021	0	AEJ: Applied Economics

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